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DRYING DEVICE AND RECORDING DEVICE

Abstract

A drying device includes a drying section that dries a medium onto which liquid was ejected by generating an electromagnetic wave in response to application of a high-frequency voltage. The drying section includes a first electrode, a second electrode arranged so as to surround the first electrode in plan view from a first direction toward a medium, a first conductor that includes a coil and that electrically connects a transmission line, which is configured to transmit a high-frequency voltage, and the first electrode, and a second conductor that electrically connects the transmission line and the second electrode. The first electrode extends in a second direction intersecting the first direction. The second electrode includes an opening section that surrounds the first electrode, and the opening section has a long side in the second direction and a short side in a third direction intersecting the first direction and the second direction. A first distance between the first electrode and the short side of the opening section in the second direction is longer than a second distance between the first electrode and the long side of the opening section in the third direction.

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Background/Summary

[0001] The present application is based on, and claims priority from JP Application Serial Number 2024-031076, filed Mar. 1, 2024, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a drying device and a recording device.

2. Related Art

[0003] For example, JP-A-2019-155811 discloses a drying device that dries a medium by generating electromagnetic waves toward a medium onto which liquid was ejected. Such a drying device generates an electromagnetic wave with respect to a medium by supplying a high-frequency voltage between a first electrode and a second electrode. By this, a medium onto which liquid was ejected can be dried.

[0004] However, in such a drying device, there is a possibility that the drying degree of a medium becomes uneven. For this reason, it is desired to equalize the drying degree of a medium.

SUMMARY

[0005] A drying device to overcome the above-described problem includes a drying section that dries a medium, onto which liquid was ejected, by generating an electromagnetic wave in response to application of a high-frequency voltage, wherein the drying section includes a first electrode, a second electrode arranged so as to surround the first electrode in plan view from a first direction toward a medium, a first conductor that includes a coil and that electrically connects a transmission line, which is configured to transmit a high-frequency voltage, and the first electrode, and a second conductor that electrically connects the transmission line and the second electrode, the first electrode extends in a second direction intersecting the first direction, the second electrode includes an opening section that surrounds the first electrode, the opening section has a long side in the second direction and a short side in a third direction intersecting the first direction and the second direction, and a first distance between the first electrode and the short side of the opening section in the second direction is longer than a second distance between the first electrode and the long side of the opening section in the third direction.

[0006] A recording device to overcome the above-described problem includes a recording section that performs recording by ejecting liquid onto a medium and a drying section that dries a medium onto which liquid was ejected by the recording section by generating an electromagnetic wave in response to application of a high-frequency voltage, wherein the drying section includes a first electrode, a second electrode arranged so as to surround the first electrode in plan view from a first direction toward a medium, a first conductor that includes a coil and that electrically connects a transmission line, which is configured to transmit a high-frequency voltage, and the first electrode, and a second conductor that electrically connects the transmission line and the second electrode, the first electrode extends in a second direction intersecting the first direction, the second electrode includes an opening section that surrounds the first electrode, the opening section has a long side in the second direction and a short side in a third direction intersecting the first direction and the second direction, and a first distance between the first electrode and the short side of the opening section in the second direction is longer than a second distance between the first electrode and the long side of the opening section in the third direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a schematic diagram showing a recording system of a first embodiment.

[0008] FIG. 2 is a perspective view showing a drying section of the first embodiment.

[0009] FIG. 3 is a schematic diagram showing the drying section of the first embodiment.

[0010] FIG. 4 is a schematic diagram showing drying sections of the first embodiment.

[0011] FIG. 5 is a perspective view showing a drying section of a second embodiment.

[0012] FIG. 6 is a schematic diagram showing a drying section of a modification.

DESCRIPTION OF EMBODIMENTS

First Embodiment

[0013] Hereinafter, an embodiment of a recording system including a drying device and a recording device will be described. In the following description, a direction intersecting a vertical direction Z is referred to as a width direction X, and a direction intersecting the vertical direction Z and the width direction X is referred to as a depth direction Y. One direction along the width direction X is referred to as a first width direction X1, and the other direction along the width direction X is referred to as a second width direction X2. One direction along the depth direction Y is referred to as a first depth direction Y1, and the other direction along the depth direction Y is referred to as a second depth direction Y2. In the vertical direction Z, an upper direction is referred to as an upper direction Z1, and a lower direction is referred to as a lower direction Z2. The vertical direction Z corresponds to an example of a first direction. The width direction X corresponds to an example of a second direction. The depth direction Y corresponds to an example of a third direction.

Configuration of Recording System 10

[0014] As shown in FIG. 1, a recording system 10 is a system that performs recording on a medium 90. In particular, the recording system 10 is a system that performs recording on the medium 90 by ejecting liquid onto the medium 90. The recording system 10 is a system that dries the medium 90 after recording onto which liquid was ejected.

[0015] The recording system 10 includes a recording device 11. The recording device 11 is configured to perform recording on the medium 90. In particular, the recording device 11 performs recording on the medium 90 by ejecting liquid onto the medium 90. The recording device 11 may be an inkjet type printer that performs recording by ejecting ink, which is an example of liquid, onto the medium 90. The medium 90 includes a front surface 90A and a back surface 90B. The medium 90 is fabric, but may be, for example, paper.

[0016] The recording system 10 includes a drying device 12. The drying device 12 is configured to dry the medium 90 after recording onto which the recording device 11 ejected liquid. In particular, the drying device 12 generates an electromagnetic wave to dry the medium 90 after recording.

[0017] The recording system 10 includes a feeding section 13. The feeding section 13 feeds the medium 90 before recording to the recording device 11. The feeding section 13 includes a feed roller 13A. The feed roller 13A extends along the width direction X. In the width direction X, the width of the feed roller 13A is longer than the width of the medium 90. The feed roller 13A is configured to rotatably hold a first roll body 91. The first roll body 91 is the medium 90 before recording that is wound and stacked. The medium 90 may be elongated. In this way, the feed roller 13A holds the medium 90 to be fed to the recording device 11.

[0018] The recording system 10 includes a winding section 14. The winding section 14 winds up the medium 90 after recording by the recording device 11. In particular, the winding section 14 winds up the medium 90 after recording and drying by the drying device 12. The winding section 14 includes a winding roller 14A. The winding roller 14A extends along the width direction X. In the width direction X, the width of the winding roller 14A is longer than the width of the medium 90. The winding roller 14A is configured to rotatably hold a second roll body 92. The second roll

body **92** is the medium **90** after recording that is wound and stacked. In this way, the winding roller **14A** winds up the medium **90** that was recorded by the recording device **11** and dried by the drying device **12**.

Configuration of Recording Device **11**

[0019] Here, the configuration of the recording device **11** will be described in detail.

[0020] The recording device **11** includes a recording section **20**, a recording support section **21**, and a recording transport section **22**. The recording section **20** is configured to perform recording on the medium **90** by ejecting liquid onto the medium **90**. The recording section **20** is configured to perform recording on the medium **90** by ejecting liquid onto the front surface **90A** of the medium **90**. The recording section **20** performs recording on the medium **90** supported by the recording support section **21**. The recording section **20** performs recording on the medium **90** transported by the recording transport section **22**.

[0021] The recording section **20** includes a head **23**. The head **23** may be a serial head or may be a line head. A serial head is a head that scans in the width direction **X** of the medium **90**. A line head is a head that records simultaneously across the width direction **X** of the medium **90**.

[0022] The head **23** includes a nozzle surface **24** in which nozzles (not shown) are opened. The nozzle surface **24** is a surface facing the lower direction **Z2**. The nozzle surface **24** is a surface facing the front surface **90A** of the medium **90** transported by the recording transport section **22**. Each of the plurality of nozzles is configured to open up in the lower direction **Z2**. Each of the plurality of nozzles is configured to eject liquid.

[0023] The recording section **20** may include a carriage **25** and a carriage support section **26**. The carriage **25** is configured to support the head **23**. The carriage support section **26** extends along the width direction **X**. The carriage support section **26** supports the carriage **25** so as to be movable along the width direction **X**. The carriage **25** is movable in the width direction **X** along the carriage support section **26** by a driving force from a drive source (not shown).

[0024] The recording support section **21** is configured to support the medium **90** transported by the recording transport section **22**. The recording support section **21** is positioned in the lower direction **Z2** of the recording section **20**. The recording support section **21** supports the back surface **90B** of the medium **90** transported by the recording transport section **22**. The recording support section **21** is positioned in the lower direction **Z2** of the head **23**.

[0025] The recording transport section **22** is configured to transport the medium **90** in a transport direction **D**. The transport direction **D** is a direction along the depth direction **Y**. The recording transport section **22** may include a plurality of rollers. Although the recording transport section **22** transports the medium **90** in the transport direction **D** using the plurality of rollers, the recording transport section **22** may transport the medium **90** in the transport direction **D** using a transport belt driven by a plurality of rollers. The recording transport section **22** may perform intermittent transport in which the transport and stop of the medium **90** are repeated.

Configuration of drying device **12**

[0026] Next, the configuration of the drying device **12** will be described in detail.

[0027] The drying device **12** includes a drying unit **30**. The drying unit **30** is configured to dry the medium **90** after recording. That is, the drying device **12** sets the medium **90** on which recording is performed by the recording section **20** as an object to be dried.

[0028] The drying unit **30** is configured to dry the medium **90** after recording by generation of electromagnetic waves. The drying unit **30** is positioned in the upper direction **Z1** of the medium **90**, but may be positioned in the lower direction **Z2** of the medium **90**, or may be positioned both in the upper direction **Z1** and in the lower direction **Z2** of the medium **90**. In this way, the vertical direction **Z** is a direction toward the medium **90**.

[0029] The drying device **12** includes a high-frequency voltage generation section **31**. The high-frequency voltage generation section **31** is configured to generate a high-frequency voltage. The high-frequency voltage generation section **31** supplies a high-frequency voltage to the drying unit

30 through a transmission line **32**.

[0030] The transmission line **32** is a line for connecting the drying unit **30** and the high-frequency voltage generation section **31**. The transmission line **32** is capable of transmitting a high-frequency voltage from the high-frequency voltage generation section **31** to the drying unit **30**. That is, the transmission line **32** is capable of transmitting a high-frequency voltage.

[0031] The transmission line **32** may be a coaxial cable, but is not limited to coaxial cable. The transmission line **32** may include a first line and a second line. The first line may be a core line of the transmission line **32**. The second line may be an electromagnetic shield that covers the first line.

[0032] The drying device **12** includes a drying transport section **33**. The drying transport section **33** is configured to transport the medium **90** in the transport direction D. The drying transport section **33** functions as an example of a movement section that relatively moves the medium **90** with respect to the drying section **36** (to be described later). The transport direction D is a direction along the depth direction Y. The transport direction D corresponds to an example of a relative movement direction.

[0033] The drying transport section **33** may transport the medium **90** in the transport direction D using a plurality of rollers. The drying transport section **33** may perform continuous transport in which the medium **90** is continuously transported. Slackening of the medium **90** may occur between the recording transport section **22** and the drying transport section **33**.

[0034] The drying device **12** includes a drying support section **34**. The drying support section **34** is configured to support the medium **90** transported by the drying transport section **33**. The drying support section **34** is positioned in the lower direction Z2 of the drying unit **30**. The drying support section **34** supports the back surface **90B** of the medium **90** transported by the drying transport section **33**. The drying support section **34** is positioned in the lower direction **22** of the drying section **36** (to be described later).

[0035] The drying device **12** includes a control section **35**. The control section **35** controls the drying device **12**. Specifically, the control section **35** controls the drying unit **30**. The control section **35** controls the high-frequency voltage generation section **31**. The control section **35** controls the drying transport section **33**.

[0036] The control section **35** may be constituted by one or more processors that execute various processes in accordance with a computer program. The control section **35** may be composed of one or more dedicated hardware circuits. The control section **35** may be configured with an application specific integrated circuit that executes at least a part of various processes. The control section **35** may be composed of a processor and a circuit including a combination of hardware circuits. The processor includes a CPU and memories such as a RAM and a ROM. The memory stores program codes or commands configured to cause the CPU to perform processes. Memory, that is computer-readable medium, includes any readable medium that can be accessed by a general-purpose or dedicated computer.

[0037] The drying unit **30** includes a drying section **36**. That is, the drying device **12** includes the drying section **36**. The drying unit **30** may include a plurality of drying sections **36**. The drying section **36** may have a rectangular shape in plan view from the vertical direction Z. The drying section **36** may be arranged so that the width direction X is a longitudinal direction. Hereinafter, plan view from the vertical direction Z is simply referred to as plan view.

[0038] The drying section **36** is configured to generate an electromagnetic wave in response to application of a high-frequency voltage. The section drying **36** generates an electromagnetic wave in response to application of a high-frequency voltage. By this, the drying section **36** is configured to dry the medium **90** onto which liquid was ejected by the recording section **20**. The drying section **36** is an electromagnetic wave generation section.

[0039] The drying section **36** generates an alternating current electric field by generating an electromagnetic wave. An electromagnetic wave generated by the drying section **36** has an electric field as a main component. The drying section **36** can significantly reduce induction of a magnetic

field due to a generated electric field as compared with an electromagnetic wave generation section that generates a normal electromagnetic wave.

[0040] As a specific example, the drying section **36** generates electromagnetic waves of 2.4 GHz, but is not limited to this. The drying section **36** may generate, for example, electromagnetic waves of 3 MHz to 300 MHz. The drying section **36**, for example, may generate electromagnetic waves of 300 MHz to 30 GHz, and among these, may generate electromagnetic waves of 10 MHz to 20 GHz.

[0041] The drying section **36** dries the medium **90** by heating the medium **90** from the front surface **90A**. Specifically, the drying section **36** heats liquid ejected onto the medium **90** from the front surface **90A**. The drying section **36** dries the medium **90** by vaporizing liquid ejected onto the medium **90**. That is, the drying section **36** is a method of drying the medium **90** regardless of whether or not water vapor is saturated around the medium **90**. Therefore, the drying section **36** does not need to blow dry gas in which water vapor is not saturated around the medium **90**.

Configuration of Drying Section **36**

[0042] As shown in FIG. **2**, the drying section **36** includes a first electrode **41**, a second electrode **42**, a first conductor **43**, and a second conductor **44**. The drying section **36** may include a facing section **45**. FIG. **2** is a view in which the first electrode **41** and second electrode **42** are arranged on a lower direction **Z2** side.

[0043] The first electrode **41** has a flat plate shape, but may have a rod shape. The first electrode **41** is elongated in the width direction **X** in plan view. That is, the first electrode **41** extends in the width direction **X** in plan view. The first electrode **41** may have a rectangular shape in plan view.

[0044] The first electrode **41** includes a first electrode surface **41A**. The first electrode surface **41A** is a surface facing the lower direction **Z2**. That is, the first electrode surface **41A** is a surface facing the front surface **90A** of the medium **90**. The first electrode **41** is arranged such that the first electrode surface **41A** abuts the facing section **45**.

[0045] The first electrode **41** includes a central section **41B** and both end sections **41C**. The central section **41B** is a section positioned at the center in the width direction **X**. Both end sections **41C** are sections positioned at both ends in the width direction **X**. The central section **41B** and both end sections **41C** are provided integrally.

[0046] The central section **41B** constitutes the first electrode surface **41A**. The central section **41B** is provided at a position overlapping with the second electrode **42** in the vertical direction **Z**. That is, at least a part of the first electrode **41** is provided at a position overlapping with the second electrode **42** in the vertical direction **Z**.

[0047] Both end sections **41C** are configured to be upwardly inclined with respect to the outside in the width direction **X**. Both end sections **41C** are positioned away from the facing section **45**. That is, both end sections **41C** extend in the upper direction **Z1** away from the medium **90** in the vertical direction **Z**. Both end sections **41C** may be curved away from the facing section **45**.

[0048] The second electrode **42** has a flat plate shape. The second electrode **42** includes a second electrode surface **42A**. The second electrode surface **42A** is a surface facing the lower direction **Z2**. That is, the second electrode surface **42A** is a surface facing the front surface **90A** of the medium **90**. The second electrode **42** is arranged such that the second electrode surface **42A** abuts the facing section **45**.

[0049] The second electrode **42** includes an opening section **42B**. The opening section **42B** has a rectangular shape in plan view, but may have a rectangular shape with rounded corners. The first electrode **41** is positioned in the opening section **42B** in plan view. The opening section **42B** surrounds the first electrode **41** in plan view. That is, the second electrode **42** is arranged so as to surround the first electrode **41** in plan view.

[0050] The first conductor **43** is configured to electrically connect the transmission line **32** and the first electrode **41**. The first conductor **43** includes a coil **43A**. The coil **43A** extends in the vertical direction **Z**. One end of the coil **43A** is connected to the first electrode **41**. The other end of the coil

43A is connected to a conductor wire **43B**.

[0051] The second conductor **44** is configured to electrically connect the transmission line **32** and the second electrode **42**. The second conductor **44** may include a columnar support **44A**. The second conductor **44** may include a plurality of columnar supports **44A**. The columnar supports **44A** are electrically connected to the second electrode **42**. The columnar support **44A** extends from the second electrode **42** in the upper direction **Z1**. The columnar support **44A** is made of metal.

[0052] The second conductor **44** may include a top plate **44B**. The top plate **44B** is electrically connected to the columnar supports **44A**. The top plate **44B** is provided at an upper end section of the columnar supports **44A**. The top plate **44B** connects a plurality of the columnar supports **44A**. The top plate **44B** may be integrated with the columnar support **44A**. The top plate **44B** is made of metal.

[0053] The facing section **45** is positioned between the first electrode **41** and the second electrode **42**, and the medium **90**. The facing section **45** may have a flat plate shape. The facing section **45** is made of a material that transmits electromagnetic waves generated by the drying section **36**. The facing section **45** is arranged so as to face the front surface **90A** of the medium **90**. The facing section **45** may not be in contact with the medium **90**, and may be in contact with the medium **90**. The facing section **45** protects the first electrode **41** and the second electrode **42**. The facing section **45** is composed of a member having insulating properties. The facing section **45** may be a glass plate. The facing section **45** may be a ceramic with high transmittance. The facing section **45** may be made of a resin with a low dissipation factor. The facing section **45** may be made of polypropylene. The facing section **45** may be made of polyethylene.

[0054] By configuring the drying section **36** in this manner, when a high-frequency voltage is applied, the first electrode **41** and the second electrode **42** heat the medium **90** by generating an electromagnetic wave in response to application of a high-frequency voltage.

[0055] Such a drying section **36** can transmit a large amount of thermal energy to the medium **90** due to generation of electromagnetic waves. The drying section **36** is not of a thermal conduction type but of an electromagnetic wave type, and does not need to include a member such as a heating wire for heating. This allows the drying section **36** to be made smaller in size.

[0056] The minimum separation distance between the first electrode **41** and the second electrode **42** is equal to or less than 1/10 of the wavelength of an electromagnetic wave output from the drying section **36**. By this, electromagnetic waves generated when a high-frequency voltage is applied can be attenuated in the vicinity of the first electrode **41** and the second electrode **42**. By this, it is possible to reduce the intensity of an electromagnetic wave that reaches a distant place from the first electrode **41** and the second electrode **42**. That is, an electromagnetic wave generated from the drying section **36** is very strong in the vicinity of the first electrode **41** and the second electrode **42**, and is very weak in a distant place.

[0057] Such a drying section **36** can intensively generate an alternating current electric field in the vicinity of the first electrode **41** and the second electrode **42** by appropriately controlling the frequency band of an electromagnetic wave to be generated. In other words, it is possible to suppress the influence on the surroundings accompanying the generation of electromagnetic waves beyond the vicinity of the first electrode **41** and the second electrode **42**. The vicinity of the first electrode **41** and the second electrode **42** may correspond to a range of, for example, 3 mm to 3 cm.

[0058] Configurations of first electrode **41** and second electrode **42**

[0059] As shown in FIG. 3, the first electrode **41** is provided so as to have the length of a distance **D11** in the width direction **X**. The second electrode **42** is provided so as to have the length of a distance **D12** in the width direction **X** and the length of a distance **D22** in the depth direction **Y**. The distance **D12** is longer than the distance **D22**.

[0060] The opening section **42B** is provided so as to have the length of a distance **D13** in the width direction **X** and the length of a distance **D23** in the depth direction **Y**. The distance **D13** is longer than the distance **D23**. In this manner, the opening section **42B** is provided such that the width

direction X corresponds to a long side **42C** and the depth direction Y corresponds to a short side **42D**. The opening section **42B** may be provided such that the distance **D23**, which is the length of the short side **42D**, is one-third or less of the distance **D13**, which is the length of the long side **42C**. The distance **D23** may be one-third or less of the distance **D11**.

[0061] A distance between the first electrode **41** and the short side **42D** in the width direction X is a first distance **D1**. A distance between the first electrode **41** and the long side **42C** in the depth direction Y is a second distance **D2**. The first distance **D1** is longer than the second distance **D2**. By configuring the first electrode **41** and the second electrode **42** in this manner, it is possible to weaken an electric field generated between both end sections **41C** and the short side **42D** of the first electrode **41**. Specifically, since both end sections **41C** of the first electrode **41** are inclined in the upper direction **Z1**, an electromagnetic wave with respect to the medium **90** can be weakened. By making the first distance **D1** longer than the second distance **D2**, it is possible to weaken an electromagnetic wave with respect to the medium **90**. By this, it is possible to suppress the concentration of a strong electric field region at both end sections **41C** of the first electrode **41** and to suppress unevenness of the drying degree of the medium **90**.

Arrangement of Drying Section **36**

[0062] As shown in FIG. **4**, when the drying unit **30** includes a plurality of the drying sections **36**, the plurality of the drying sections **36** may be arranged in the width direction X, and the plurality of the drying sections **36** may be arranged in the depth direction Y. In this case, a plurality of the drying sections **36** may be arranged across the entire width of the medium **90** in the width direction X. In this manner, a plurality of the drying sections **36** is arranged so as to be aligned in the width direction X. A plurality of the drying sections **36** is arranged so as to be aligned in the depth direction Y.

[0063] A plurality of the drying sections **36** includes first drying sections **36A** and second drying sections **36B**. The first drying section **36A** and the second drying section **36B** may generate an electromagnetic wave having different intensities. The first drying section **36A** generates an stronger electromagnetic wave than the second drying section **36B**, but may also generate a weaker electromagnetic wave than the second drying section **36B**.

[0064] Specifically, by configuring the inductance of the coil **43A** of the first drying section **36A** to be different from the inductance of the coil **43A** of the second drying section **36B**, the resonance frequency of the first drying section **36A** can be made different from the resonance frequency of the second drying section **36B**.

[0065] As described above, the first drying section **36A** and the second drying section **36B** have different resonance frequencies. As a specific example, a plurality of the first drying sections **36A** in a first row **C1** and a plurality of second drying sections **36B** in a second row **C2** have different resonance frequencies.

[0066] The first drying section **36A** is arranged in a first region **R1**. The second drying section **36B** is arranged in a second region **R2**. The first region **R1** is a region positioned on an upstream side in the transport direction **D**. The second region **R2** is a region positioned on a downstream side in the transport direction **D**. That is, the second region **R2** is a region positioned on a downstream side of the first region **R1** in the transport direction **D**.

[0067] In the first region **R1**, the drying sections **36** are arranged in the same number of rows as in the second region **R2** in the depth direction Y. The number of rows of the drying sections **36** arranged in the first region **R1** in the depth direction Y may be smaller than that in the second region **R2**. The number of rows of the drying sections **36** arranged in the first region **R1** in the depth direction Y may be larger than that in the second region **R2**.

[0068] The plurality of the drying sections **36** is arranged so as to be aligned in the width direction X in each row. As a specific example, a plurality of the drying sections **36** is arranged so as to be aligned in the width direction X in the first row **C1**. A plurality of the drying sections **36** is arranged so as to be aligned in the width direction X in the second row **C2**. The first row **C1** and the second

row C2 may be rows that are adjacent to each other in the depth direction Y.

[0069] In this way, a plurality of the drying sections 36 in the first row C1 and a plurality of the drying sections 36 in the second row C2 is arranged so as to be aligned in the depth direction Y. A plurality of the drying sections 36 in the first row C1 and a plurality of the drying sections 36 in the second row C2 is arranged such that their positions in the width direction X are different.

[0070] As a specific example, a case where the medium 90 on which pigment ink is ejected as liquid is dried will be described. Pigment ink includes a pigment, water, and a solvent. The solvent may include, for example, glycerin. Glycerin is a solvent for preventing clogging of a nozzle that ejects liquid. Due to the vaporization of glycerin, the abrasion resistance of the medium 90 onto which pigment ink was ejected is improved.

[0071] When drying the medium 90 onto which pigment ink was ejected, provisional drying and main drying are performed. Provisional drying vaporizes water contained in pigment ink at approximately 100° C. In such a case, it is desirable to rapidly increase the temperature of pigment ink.

[0072] Main drying is performed after provisional drying. Main drying vaporizes a solvent contained in pigment ink at approximately 300° C. In main drying, it is desirable to maintain the temperature of pigment ink without excessively increasing the temperature in order to increase the reliability of vaporization of glycerin without causing thermal denaturation of the medium 90. It is desirable that main drying is performed for a longer time than provisional drying.

[0073] For this reason, the first drying section 36A generates an electromagnetic wave that are stronger than that of the second drying section 36B. In the first region R1, the drying sections 36 are arranged in a smaller number of rows in the depth direction Y than in the second region R2.

[0074] In provisional drying, the first drying section 36A generates an electromagnetic wave stronger than that of the second drying section 36B, so that the temperature can be rapidly increased. Thereafter, in main drying, the temperature of the second drying section 36B can be maintained without being excessively increased.

[0075] The medium 90 is transported in the transport direction D at a constant speed. Therefore, in a case where the number of rows of the drying sections 36 arranged in the first region R1 is smaller than the number of rows of the drying sections 36 arranged in the second region R2, it is possible to make the time for drying by the second drying section 36B arranged in the second region R2 longer than the time for drying by the first drying section 36A arranged in the first region R1.

[0076] For example, as the medium 90 dries, the amount of water contained in medium 90 decreases. As described above, when the water content of the medium 90 containing water having a high dielectric constant decreases, the relative dielectric constant decreases. Therefore, it is desirable that the resonance frequency is changed stepwise in the transport direction D.

Operations and Effects of First Embodiment

[0077] Operations and effects of the first embodiment will be described.

[0078] (1-1) In the related art, the first electrode extends in the width direction X, and a strong electric field region tends to be concentrated in nearby regions of both end sections of the first electrode, for example, by the same principle as that of corona discharge. Therefore, the heating amount is large in the nearby regions of both end sections of the first electrode, but the heating amount is small in the other region. By this, since the heating amount becomes uneven depending on a region, the drying degree of the medium 90 becomes uneven.

[0079] Therefore, the first distance D1 between the first electrode 41 and the short side 42D of the opening section 42B in the width direction X is longer than the second distance D2 between the first electrode 41 and the long side 42C of the opening section 42B in the depth direction Y.

According to this configuration, it is possible to avoid the concentration of electromagnetic waves at both end sections 41C of the first electrode 41. By this, electromagnetic waves to be generated can be equalized. Therefore, unevenness in the drying degree of the medium 90 can be suppressed.

[0080] (1-2) A plurality of the drying sections 36 is arranged so as to be aligned in the depth

direction Y. According to this configuration, by arranging the plurality of the drying sections **36** so as to be aligned in the depth direction Y, it is possible to suppress a decrease in the drying efficiency of the medium **90**.

[0081] (1-3) A plurality of the drying sections **36** is arranged so as to be aligned in the width direction X. According to this configuration, by arranging the plurality of the drying sections **36** so as to be aligned in the width direction X, it is possible to suppress a decrease in the drying efficiency of the medium **90**.

[0082] (1-4) A plurality of the drying sections **36** is arranged so as to be aligned in the width direction X in the first row **C1**, and a plurality of the drying sections **36** is arranged so as to be aligned in the width direction X in the second row **C2** adjacent to the first row **C1** in the depth direction Y. A plurality of the drying sections **36** in the first row **C1** and a plurality of the drying sections **36** in the second row **C2** is arranged such that their positions in the width direction X are different. According to this configuration, by arranging the plurality of the drying sections **36** so as to be aligned in both the width direction X and the depth direction Y, it is possible to suppress a decrease in the drying efficiency of the medium **90**. In addition, since the drying sections **36** in the first row **C1** and the drying sections **36** in the second row **C2** are arranged at different positions in the width direction X, it is possible to suppress unevenness of the drying degree of the medium **90**.

[0083] (1-5) A part of the first electrode **41** is provided at a position overlapping with the second electrode **42** in the vertical direction Z. According to this configuration, both the first electrode **41** and the second electrode **42** can have the same distance from the medium **90** in the vertical direction Z. Therefore, a decrease in the drying efficiency of the medium **90** can be suppressed.

[0084] (1-6) The first electrode **41** includes the central section **41B** in the width direction X and both end sections **41C** in the width direction X. Both end sections **41C** extend in the upper direction **Z1** away from the medium **90** in the vertical direction Z. According to this configuration, it is possible to avoid the concentration of electromagnetic waves at both end sections **41C** of the first electrode **41**. By this, electromagnetic waves to be generated can be equalized. Therefore, unevenness in the drying degree of the medium **90** can be suppressed.

[0085] (1-7) The central section **41B** and both end sections **41C** are provided integrally. According to this configuration, since the central section **41B** and both end sections **41C** are provided integrally, a high-frequency voltage can be efficiently transmitted. Therefore, a decrease in the drying efficiency of the medium **90** can be suppressed.

[0086] (1-8) The drying transport section **33** relatively moves the medium **90** along the depth direction Y with respect to the drying section **36**. According to this configuration, it is possible to relatively move the medium **90** with respect to the drying section **36** along the depth direction Y, which intersects the width direction X that is the longitudinal direction of the first electrode **41** and the opening section **42B**. By this, a region to be dried can be widened in the width direction X that does not depend on the relative movement of the medium **90** with respect to the drying section **36**. Therefore, unevenness in the drying degree of the medium **90** can be suppressed.

[0087] (1-9) A plurality of the first drying sections **36A** is arranged in the first row **C1** so as to be aligned in the width direction X. A plurality of the second drying sections **36B** is arranged in the second row **C2** so as to be aligned in the width direction X. The plurality of the first drying sections **36A** in the first row **C1** and the plurality of the second drying sections **36B** in the second row **C2** have different resonance frequencies. According to this configuration, resonance frequencies can be made different between the plurality of the first drying sections **36A** in the first row **C1** and the plurality of the second drying sections **36B** in the second row **C2** arranged in the depth direction Y in which the medium **90** relatively moves with respect to the drying section **36**. Therefore, the drying degree of the medium **90** can be adjusted.

Second Embodiment

[0088] Next, a second embodiment will be described. In the following description, the same configuration as that of the embodiment already described will be omitted or simplified, and a

configuration different from that of the embodiment already described will be described.

[0089] As shown in FIG. 5, in the second embodiment, the second electrode **42** includes a plurality of the opening sections **42B**. In the width direction X, the second electrode **42** may be longer than the width of the medium **90**. In the width direction X, the long side **42C** of the opening section **42B** may be longer than the width of the medium **90**.

[0090] The drying section **36** includes a plurality of first electrodes **41** and the first conductors **43**. Each of the plurality of the first electrodes **41** is arranged so as to be surrounded by the plurality of the opening sections **42B**. In the width direction X, each of the plurality of the first electrodes **41** may be longer than the width of the medium **90**. The distances between the first electrode **41** and the short side **42D** in the width direction X are longer than the distances between the first electrode **41** and the long side **42C** in the depth direction Y.

[0091] The drying transport section **33** relatively moves the medium **90** along the depth direction Y with respect to the drying section **36**. The relative movement direction is a direction along the depth direction Y. That is, the width direction X intersecting the transport direction D is a longitudinal direction of the drying section **36**.

[0092] Thus, the drying section **36** according to the second embodiment is configured to include the plurality of the drying sections **36** according to the first embodiment. Specifically, the drying section **36** according to the second embodiment is configured such that the first electrode **41** and the second electrode **42** are elongated in the width direction X. In the drying section **36** according to the second embodiment, a plurality of the second electrodes **42** arranged in the depth direction Y according to the first embodiment are integrally configured.

Operations and Effects of Second Embodiment

[0093] Operations and effects of the second embodiment will be described.

[0094] (2-1) In the width direction X, the long sides **42C** of the opening sections **42B** are longer than the width of the medium **90**, and the first electrodes **41** are longer than the width of the medium **90**. According to this configuration, it is possible to relatively move the medium **90** with respect to the drying section **36** along the depth direction Y, which intersects the width direction X that is the longitudinal direction of the first electrode **41** and the opening section **42B**. By this, a region to be dried can be widened in the width direction X that does not depend on the relative movement of the medium **90** with respect to the drying section **36**. Therefore, unevenness in the drying degree of the medium **90** can be suppressed.

Modifications

[0095] The present embodiment can be implemented with the following modifications. The present embodiment and the following modifications can be implemented in combination with each other within a technically compatible range.

[0096] In the second embodiment, the drying transport section **33** may relatively move the medium **90** along the width direction X with respect to the drying section **36**. The relative movement direction is a direction along the width direction X. That is, the width direction X along the transport direction D is a longitudinal direction of the drying section **36**.

[0097] In the second embodiment, as shown in FIG. 6, in a case where the drying transport section **33** relatively moves the medium **90** along the depth direction Y with respect to the drying section **36**, a direction inclined with respect to the width direction X and the depth direction Y may be a longitudinal direction of the drying section **36**. The relative movement direction is a direction inclined with respect to the width direction X and the depth direction Y.

[0098] As described above, the relative movement direction in which the medium **90** is relatively moved with respect to the drying section **36** is a direction inclined with respect to the width direction X. According to this configuration, even in a case where the medium **90** is relatively moved in the relative movement direction inclined with respect to the width direction X, it is possible to continuously dry the medium **90**. Therefore, unevenness in the drying degree of the medium **90** can be suppressed.

[0099] The drying device **12** may include a movement section that moves the drying section **36**. The drying device **12** may include a movement section that moves both the drying section **36** and the medium **90**. In this manner, a movement section may be configured to relatively move the medium **90** with respect to the drying section **36**.

[0100] In the first embodiment, the plurality of the first drying sections **36A** in the first row **C1** and the plurality of the second drying sections **36B** in the second row **C2** may not be adjacent to each other in the depth direction **Y**. That is, in the depth direction **Y**, the drying section **36** in another row may be arranged between the plurality of the first drying sections **36A** in the first row **C1** and the plurality of the second drying sections **36B** in the second row **C2**.

[0101] The first electrode **41** may be inclined in the upper direction **Z1** so as to branch both end sections **41C**. Both end sections **41C** of the first electrode **41** need not be inclined in the upper direction **Z1**. That is, both end sections **41C** may be provided at the same positions as the central section **41B** in the vertical direction **Z**. In this way, the first electrode **41** is provided at a position overlapping with the second electrode **42** in the vertical direction **Z**.

[0102] The drying section **36** may be arranged so that the depth direction **Y** is a longitudinal direction. That is, the depth direction **Y** may be an example of the second direction. The width direction **X** may be an example of the third direction. The drying section **36** may be arranged so as to be inclined with respect to the width direction **X** and the depth direction **Y**. A direction inclined with respect to the width direction **X** and the depth direction **Y** may be an example of the second direction.

[0103] The drying unit **30** may include one or more drying sections **36**. A plurality of the drying sections **36** may be arranged in a plurality of rows in the width direction **X**. A plurality of the drying sections **36** may be arranged in a plurality of rows in the depth direction **Y**.

[0104] The drying section **36** may be provided on a back surface **90B** side of the medium **90**. The drying section **36** may be provided on both a front surface **90A** side of the medium **90** and a back surface **90B** side of the medium **90**. The drying section **36** may be capable of scanning in the width direction **X**.

[0105] The drying section **36** may be provided separately from the facing section **45**. That is, the drying section **36** may not include the facing section **45**. In this case, it is desirable for the facing section **45** to be provided between the first electrode **41** and the second electrode **42**, and the medium **90**.

[0106] The first electrode **41** is not limited to a flat plate shape and may, for example, have a substantially flat plate shape. The substantially flat shape may include, for example, a shape curved in the thickness direction, which is a direction along the vertical direction **Z**, or a linear shape with an extremely large aspect ratio of a rectangular shape.

[0107] The second electrode **42** is not limited to a flat plate shape and may, for example, have a substantially flat plate shape. The substantially flat shape may include, for example, a shape curved in the thickness direction, which is a direction along the vertical direction **Z**, or a linear shape with an extremely large aspect ratio of a rectangular shape.

[0108] At least one of the first electrode surface **41A** and the second electrode surface **42A** is not limited to a planar shape and may be a substantially planar shape. The substantially planar shape may include, for example, a shape curved in the thickness direction, which is a direction along the vertical direction **Z**, or a linear shape with an extremely large aspect ratio of a rectangular shape.

[0109] The drying section **36** may not be provided in the drying device **12** and may be provided in the recording device **11**. That is, the recording device **11** may include the drying section **36**. In this case, it is sufficient that the drying section **36** is provided on a downstream side of the recording section **20** in the transport direction **D**. In this manner, the drying section **36** may be applied to the recording device **11** instead of the drying device **12**.

[0110] A lateral type printer may be adopted as the recording device **11**. The lateral type printer is a printer in which the carriage **25** can move in two directions, a main scanning direction and a sub-

scanning direction.

[0111] The medium **90** is not limited to a roll body. The medium **90** may be a paper sheet, a resin film or sheet, a resin-metal composite film, a laminate film, a textile, a nonwoven fabric, a metal foil, a metal film, a ceramic sheet, a garment, or the like.

[0112] Liquid can be arbitrarily selected as long as it can perform recording on the medium **90** by depositing to medium **90**. For example, ink includes ink in which particles of a functional material made of a solid material such as pigment or a metal particle are dissolved, dispersed, or mixed in a solvent, and includes various compositions such as water-based ink, oil-based ink, gel ink, and hot melt ink.

[0113] As used herein, the phrase “at least any” means one or more of the desired options. As an example, the phrase “at least any” as used herein means only one option if the number of options is two, or both of the two options. As another example, the phrase “at least any” as used herein means only one option or a combination of any two or more options when the number of options is three or more.

Notes

[0114] Hereinafter, technical ideas grasped from the above-described embodiments and modifications, and operations and effects thereof will be described. The present technical ideas and the operations and effects thereof can be combined with each other within a technically consistent range.

[0115] (A) A drying device includes a drying section that dries a medium, onto which liquid was ejected, by generating an electromagnetic wave in response to application of a high-frequency voltage, wherein the drying section includes a first electrode, a second electrode arranged so as to surround the first electrode in plan view from a first direction toward a medium, a first conductor that includes a coil and that electrically connects a transmission line, which is configured to transmit a high-frequency voltage, and the first electrode, and a second conductor that electrically connects the transmission line and the second electrode, the first electrode extends in a second direction intersecting the first direction, the second electrode includes an opening section that surrounds the first electrode, the opening section has a long side in the second direction and a short side in a third direction intersecting the first direction and the second direction, and a first distance between the first electrode and the short side of the opening section in the second direction is longer than a second distance between the first electrode and the long side of the opening section in the third direction.

[0116] According to this configuration, the first electrode extends in the second direction, and the first distance between the first electrode and the short side of the opening section in the second direction is longer than the second distance between the first electrode and the long side of the opening section in the third direction. Therefore, it is possible to avoid the concentration of electromagnetic waves at both end sections of the first electrode. By this, electromagnetic waves to be generated can be equalized. Therefore, unevenness in the drying degree of the medium can be suppressed.

[0117] (B) The above-described drying device may be configured such that a plurality of the drying sections is arranged so as to be aligned in the third direction.

[0118] According to this configuration, by arranging the plurality of the drying sections so as to be aligned in the third direction, it is possible to suppress a decrease in the drying efficiency of the medium.

[0119] (C) The above-described drying device may be configured such that a plurality of the drying sections is arranged so as to be aligned in the second direction.

[0120] According to this configuration, by arranging the plurality of the drying sections so as to be aligned in the second direction, it is possible to suppress a decrease in the drying efficiency of the medium.

[0121] (D) The above-described drying device may be configured such that a plurality of the

drying sections is arranged so as to be aligned in the second direction in a first row, a plurality of the drying sections is arranged so as to be aligned in the second direction in a second row adjacent to the first row in the third direction, and the plurality of the drying sections in the first row and the plurality of the drying sections in the second row are arranged at different positions in the second direction.

[0122] According to this configuration, by arranging the plurality of the drying sections so as to be aligned in both the second direction and the third direction, it is possible to suppress a decrease in the drying efficiency of the medium. In addition, since the plurality of the drying sections in the first row and the plurality of the drying sections in the second row are arranged at different positions in the second direction, it is possible to suppress unevenness in the drying degree of the medium.

[0123] (E) The above-described drying device may be configured such that the opening section is provided so that a length of the short side is one-third or less of a length of the long side.

[0124] According to this configuration, it is possible to achieve the same effect as (A).

[0125] (F) The above-described drying device may be configured such that the first electrode is provided at a position overlapping with the second electrode in the first direction.

[0126] According to this configuration, both the first

[0127] electrode and the second electrode can have the same distance from the medium in the first direction. Therefore, it is possible to suppress a decrease in the drying efficiency of the medium.

[0128] (G) The above-described drying device may be configured such that the first electrode includes a central section in the second direction and both end sections in the second direction and the both end sections extend in a direction away from a medium in the first direction.

[0129] According to this configuration, it is possible to avoid the concentration of electromagnetic waves at both end sections of the first electrode. By this, electromagnetic waves to be generated can be equalized. Therefore, unevenness in the drying degree of the medium can be suppressed.

[0130] (H) The above-described drying device may be configured such that the central section and the both end sections are provided integrally.

[0131] According to this configuration, since the central section and both end sections are provided integrally, a high-frequency voltage can be efficiently transmitted. Therefore, it is possible to suppress a decrease in the drying efficiency of the medium.

[0132] (I) The above-described drying device may be configured such that the drying device further includes a movement section that relatively moves a medium along the third direction with respect to the drying section.

[0133] According to this configuration, it is possible to relatively move the medium with respect to the drying section along the third direction which intersects the second direction, which is a longitudinal direction of the first electrode and the opening section. By this, it is possible to widen a region to be dried in the second direction, which does not depend on the medium being relatively moved with respect to the drying section. Therefore, unevenness in the drying degree of the medium can be suppressed.

[0134] (J) The above-described drying device may be configured such that a plurality of the drying sections is arranged so as to be aligned in the second direction in a first row, a plurality of the drying sections is arranged so as to be aligned in the second direction in a second row, and the plurality of the drying sections in the first row and the plurality of the drying sections in the second row have different resonance frequencies.

[0135] According to this configuration, the resonance frequencies can be made different between the plurality of the drying sections in the first row and the plurality of the drying sections in the second row, which are arranged in the third direction in which the medium relatively moves with respect to the drying sections. Therefore, the drying degree of the medium can be adjusted.

[0136] (K) The above-described drying device may be configured such that the drying device further includes a movement section that relatively moves a medium along a relative movement

direction with respect to the drying section, wherein the relative movement direction is a direction that has an inclination with respect to the second direction.

[0137] According to this configuration, even when the medium is moved in the relative movement direction, the medium can be continuously dried. Therefore, unevenness in the drying degree of the medium can be suppressed.

[0138] (L) A recording device includes a recording section that performs recording by ejecting liquid onto a medium and a drying section that dries a medium onto which liquid was ejected by the recording section by generating an electromagnetic wave in response to application of a high-frequency voltage, wherein the drying section includes a first electrode, a second electrode arranged so as to surround the first electrode in plan view from a first direction toward a medium, a first conductor that includes a coil and that electrically connects a transmission line, which is configured to transmit a high-frequency voltage, and the first electrode, and a second conductor that electrically connects the transmission line and the second electrode, the first electrode extends in a second direction intersecting the first direction, the second electrode includes an opening section that surrounds the first electrode, the opening section has a long side in the second direction and a short side in a third direction intersecting the first direction and the second direction, and a first distance between the first electrode and the short side of the opening section in the second direction is longer than a second distance between the first electrode and the long side of the opening section in the third direction.

[0139] According to this configuration, it is possible to achieve the same effect as (A).

Claims

1. A drying device comprising: a drying section that dries a medium, onto which liquid was ejected, by generating an electromagnetic wave in response to application of a high-frequency voltage, wherein the drying section includes a first electrode, a second electrode arranged so as to surround the first electrode in plan view from a first direction toward a medium, a first conductor that includes a coil and that electrically connects a transmission line, which is configured to transmit a high-frequency voltage, and the first electrode, and a second conductor that electrically connects the transmission line and the second electrode, the first electrode extends in a second direction intersecting the first direction, the second electrode includes an opening section that surrounds the first electrode, the opening section has a long side in the second direction and a short side in a third direction intersecting the first direction and the second direction, and a first distance between the first electrode and the short side of the opening section in the second direction is longer than a second distance between the first electrode and the long side of the opening section in the third direction.
2. The drying device according to claim 1, wherein a plurality of the drying sections is arranged so as to be aligned in the third direction.
3. The drying device according to claim 1, wherein a plurality of the drying sections is arranged so as to be aligned in the second direction.
4. The drying device according to claim 1, wherein a plurality of the drying sections is arranged so as to be aligned in the second direction in a first row, a plurality of the drying sections is arranged so as to be aligned in the second direction in a second row adjacent to the first row in the third direction, and the plurality of the drying sections in the first row and the plurality of the drying sections in the second row are arranged at different positions in the second direction.
5. The drying device according to claim 1, wherein the opening section is provided so that a length of the short side is one-third or less of a length of the long side.
6. The drying device according to claim 1, wherein the first electrode is provided at a position overlapping with the second electrode in the first direction.
7. The drying device according to claim 1, wherein the first electrode includes a central section in

the second direction and both end sections in the second direction and the both end sections extend in a direction away from a medium in the first direction.

8. The drying device according to claim 7, wherein the central section and the both end sections are provided integrally.

9. The drying device according to claim 1, further comprising: a movement section that relatively moves a medium along the third direction with respect to the drying section.

10. The drying device according to claim 9, wherein a plurality of the drying sections is arranged so as to be aligned in the second direction in a first row, a plurality of the drying sections is arranged so as to be aligned in the second direction in a second row, and the plurality of the drying sections in the first row and the plurality of the drying sections in the second row have different resonance frequencies.

11. The drying device according to claim 1, further comprising: a movement section that relatively moves a medium along a relative movement direction with respect to the drying section, wherein the relative movement direction is a direction that has an inclination with respect to the second direction.

12. A recording device comprising: a recording section that performs recording by ejecting liquid onto a medium and a drying section that dries a medium onto which liquid was ejected by the recording section by generating an electromagnetic wave in response to application of a high-frequency voltage, wherein the drying section includes a first electrode, a second electrode arranged so as to surround the first electrode in plan view from a first direction toward a medium, a first conductor that includes a coil and that electrically connects a transmission line, which is configured to transmit a high-frequency voltage, and the first electrode, and a second conductor that electrically connects the transmission line and the second electrode, the first electrode extends in a second direction intersecting the first direction, the second electrode includes an opening section that surrounds the first electrode, the opening section has a long side in the second direction and a short side in a third direction intersecting the first direction and the second direction, and a first distance between the first electrode and the short side of the opening section in the second direction is longer than a second distance between the first electrode and the long side of the opening section in the third direction.
